A Survey of AI for Water Environmental Systems

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***Abstract*—Artificial intelligence (AI) has expanded its applications in water environmental systems, including urban drainage systems, wastewater treatment plants, natural aquatic systems, and constructed wetlands. Driven by advances in sensing technologies, the internet of things (IoT), and edge computing, water environmental management is shifting from experience-based practices toward data-driven decisions. Unlike existing reviews that focus on individual tasks or specific application scenarios, this review adopts a system-level perspective to examine AI applications across five core tasks—predictive modeling, pollution identification, intelligent control, multi-objective optimization, and knowledge discovery—within four water environmental systems. Attention is paid to the distribution of AI methods across tasks and system types. Based on a synthesis of literature statistics, this review summarizes the evolution of AI applications from prediction-oriented studies toward approaches emphasizing system control and optimization. It highlights closed-loop control, coordinated multi-objective optimization, and engineering-level edge deployment as key directions for future research.**

***Index Terms*—Artificial Intelligence, Deep Learning, Water Environmental Systems, Urban Drainage Systems, Wastewater Treatment Plants, Natural Aquatic Systems, Constructed Wetlands, Predictive Modeling, Pollution Identification, Intelligent Control, Multi-objective Optimization, Knowledge Discovery**

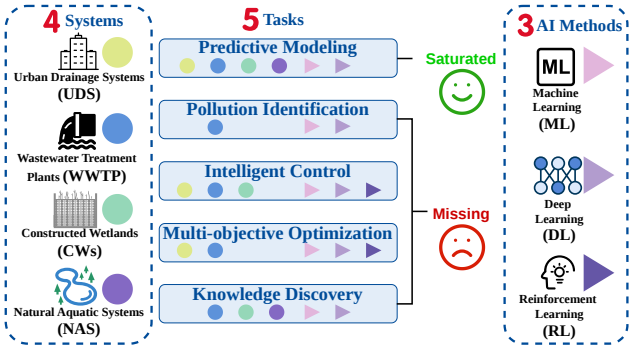

Fig. 1. **Color-coded mapping of artificial intelligence tasks and method categories across water environmental systems.** The figure illustrates the relationships among four representative water environmental systems, five core AI tasks, and three major AI method categories. Colored circular markers denote task–system associations, with consistent colors indicating the corresponding system in which the task has been investigated. Triangular markers represent AI method categories (machine learning, deep learning, and reinforcement learning), with different colors distinguishing method types and indicating their associations with specific tasks. The absence of a marker suggests limited or no reported AI applications for the corresponding task–system or task–method combination.

I. Introduction

Water environmental systems are complex systems formed by the interplay of natural processes, engineered infrastructure, and human activities, whose operational states are influenced by multiple factors, including hydrological conditions, pollutant loads, biological processes, and management strategies. With accelerated urbanization and intensified industrial activities, these systems face increasing challenges in water quality safety, resource utilization efficiency, and operational stability, particularly in representative scenarios such as wastewater treatment, urban drainage, protection of natural water bodies, and constructed wetlands. The operation of such systems exhibits pronounced nonlinearity, uncertainty, and multi-objective characteristics [1], and traditional empirical or mechanistic approaches increasingly show limitations in handling complex operating conditions, multi-source data integration, and real-time decision-making.

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To track the latest advances, we maintain a repository at https://github.com/chuny9743/AI4WaterEnv.

Against this background, artificial intelligence methods, leveraging their advantages in nonlinear modeling, feature extraction, and complex system modeling, have gradually been introduced into the water environmental systems domain to support key tasks such as water quality prediction, system operation optimization, intelligent control, and decision support. With the deployment of sensor networks, the development of online monitoring technologies, and the enhancement of computational resources, the scale and variety of data available in water environment systems continue to expand, further promoting the application of machine learning and deep learning methods across diverse water environment systems. However, these systems are typically characterized by multi-process coupling, concurrent multiple tasks, and multi-objective constraints [1], and significant differences exist across systems in terms of data structure, control requirements, and operational mechanisms [2], which leads to a highly scenario-specific and task-oriented application of artificial intelligence methods.

Previous studies have provided relatively systematic reviews of artificial intelligence applications in the water environment systems at the level of individual systems or specific tasks. However, the existing literature largely focuses on a single environmental system or a single task type. In the context of wastewater treatment systems, some studies have examined global research progress and trends in machine learning applications development from a bibliographic perspective [3],

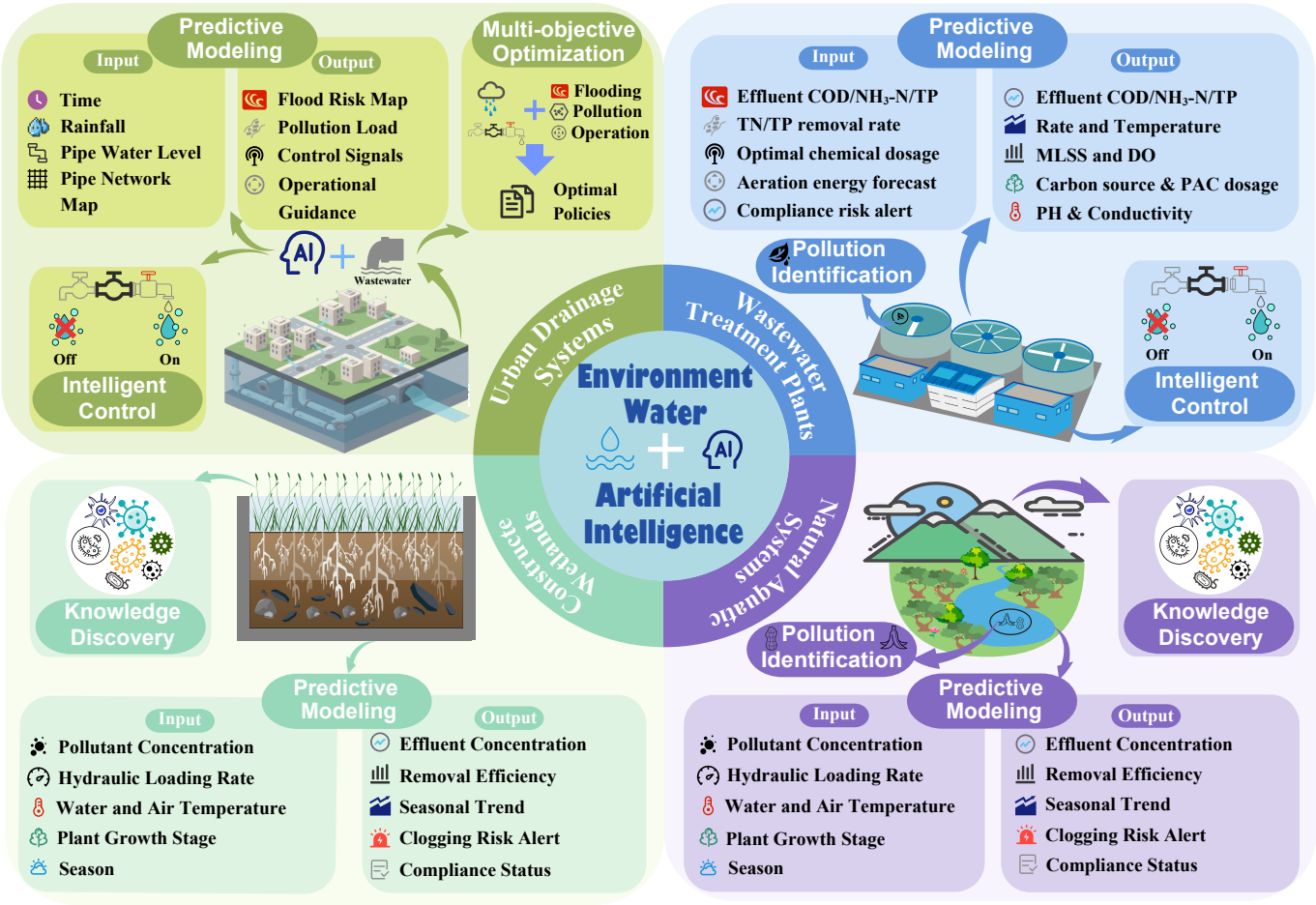

Fig. 2. **Overview of artificial intelligence (AI) applications in water environmental systems.** The four surrounding boxes correspond to urban drainage systems, wastewater treatment plants, natural aquatic systems, and constructed wetlands. Within each box, the major AI tasks widely investigated in the corresponding system are summarized, illustrating how different water environmental systems emphasize distinct task combinations.

while others have discussed the potential and challenges of artificial intelligence in wastewater treatment from technical, economic, and managerial perspectives [4]. However, these reviews are mainly limited to wastewater treatment as a single water environmental system. At the task level, several reviews have focused on prediction problems, summarizing the evolution of deep learning approaches for predicting water environments and highlighting their advantages in addressing data scarcity and incompleteness [5]. At the system level, related issues in urban water system management have also been reviewed [6]–[8].

In addition, some reviews have summarized the application of artificial intelligence from a broader perspective of water resources or natural water systems. For example, certain studies [9] have mainly concentrated on deep learning methods in hydrological and water resources systems, with relatively limited attention to water environment systems characterized by strong engineering attributes and operational constraints. Other reviews [10], [11] have focused on water quality management or prediction tasks in natural water systems, providing overviews of artificial intelligence applications in specific scenarios. Overall, existing reviews tend to concentrate on a single environmental system or a single task dimension, and a unified and systematic synthesis that examines commonalities and differences across diverse water environment systems, as well as the role of artificial intelligence under multi-task and multi-objective settings, remains limited.

From a task-oriented perspective, artificial intelligence applications in water environmental systems can be categorized into five core tasks: predictive modeling, pollution identification, intelligent control, multi-objective optimization, and knowledge discovery. However, the distribution of these tasks across different types of water environmental systems is markedly uneven. As illustrated in Fig. 1, certain AI tasks have been extensively studied in specific systems, while their applications in other systems remain limited or even absent. Predictive modeling has been widely adopted across all four types of water environmental systems. In contrast, intelligent control and multi-objective optimization are mainly concentrated in engineering-oriented systems such as urban drainage systems and wastewater treatment plants. In contrast, studies related to pollution identification and knowledge discovery are relatively scarce in some system contexts.

To provide a unified conceptual perspective, Fig. 2 presents an overview of AI applications in water environmental systems, with a focus on the integration of artificial intelligence

and water environmental systems. The figure summarizes the major AI tasks that have been investigated within urban drainage systems, wastewater treatment plants, natural aquatic systems, and constructed wetlands. It highlights cross-system imbalances in AI application patterns and supports an intuitive cross-task comparison.

In a broader sustainability context, water environmental systems are also closely coupled with energy consumption, food production, and climate dynamics, suggesting that the task–system integration discussed in this review could be further extended toward the water–energy–food–climate nexus in future research.

In response to the aforementioned research gaps, this review systematically examines artificial intelligence applications across four representative water environmental systems, namely urban drainage systems (UDS, section III-B), wastewater treatment plants (WWTP, section III-C), natural aquatic systems (NAS, section III-D, and constructed wetlands (CWs,section III-E). Within a unified framework, five core AI tasks are discussed, including predictive modeling (section IV-A), pollution identification (section IV-B), intelligent control (section IV-C), multi-objective optimization (section IV-D), and knowledge discovery (section IV-E). By jointly considering system types and task categories, this review aims to provide an integrated perspective on the role of artificial intelligence in the analysis, operation, and management of water environmental systems.

II. Background

A. Early Development of AI in Water Environmental Systems

The application of artificial intelligence in water environmental systems can be traced back to early studies [107] [108] [109] [110] in the late twentieth century, when rule-based expert systems and shallow artificial neural networks were introduced to support data analysis and process understanding. During this initial stage, AI techniques mainly served as auxiliary tools complementing traditional statistical and process-based models, rather than acting as core components of system operation or control.

Early applications were largely concentrated in Predictive Modeling tasks, such as water quality forecasting, hydrological variable estimation, and system state prediction in natural aquatic systems and wastewater treatment plants. These studies focused on capturing nonlinear relationships between input variables and system responses under relatively limited data availability. Intelligent control and multi-objective optimization were rarely addressed during this period, mainly due to constraints in sensing infrastructure, computational resources, and real-time data acquisition.

The historical timeline of AI-related studies in water environmental systems illustrates a gradual increase in research activity from the 1990s onward, followed by a more pronounced growth after the mid-2010s. This growth coincides with advances in machine learning algorithms, increased data availability, and improvements in computational capabilities, laying the foundation for broader AI adoption across different water environmental systems.

B. Global Distribution of AI in Water Environmental Systems

In addition to temporal evolution, AI applications in water environmental systems exhibit distinct spatial patterns. The global distribution of AI-related studies, illustrated in Fig. 3, indicates that research activity is unevenly distributed across regions and system types.

Studies focusing on wastewater treatment plants and urban drainage systems are predominantly reported in regions with well-established water infrastructure and long-term monitoring programs, where high-quality operational data are more readily available. In contrast, research on natural aquatic systems spans a wider geographical range, reflecting their global relevance to water quality management and ecosystem protection. However, AI applications in constructed wetlands remain comparatively limited in many regions, partly due to site-specific system configurations, heterogeneous operating conditions, and relatively sparse monitoring data.

Overall, the spatial distribution of AI applications suggests that data availability, system controllability, and management priorities play a critical role in shaping research focus across different water environmental systems.

C. Recent Trends in System- and Task-Oriented AI Research

Recent literature from 2021 to 2025 reveals a noticeable shift in both system focus and task orientation of AI applications in water environmental systems. As summarized in Fig. 2, predictive modeling remains the most extensively studied task across all four system types. However, an increasing number of studies have begun to address intelligent control and multi-objective optimization, particularly in urban drainage systems and wastewater treatment plants.

The statistical distribution of recent studies indicates that engineered water environmental systems are increasingly used as platforms for AI-driven operational decision making, including real-time control and coordinated optimization. In contrast, natural aquatic systems and constructed wetlands continue to be dominated by predictive modeling and pollution identification tasks, reflecting inherent differences in system objectives, controllability, and operational constraints.

At the task level, knowledge discovery has emerged as an important complementary direction, supporting data-driven understanding of system behavior and latent process interactions. Nevertheless, its application remains uneven across system types and is often integrated with predictive modeling rather than treated as an independent task.

D. From Prediction to System-Level Intelligence

Taken together, the temporal trends shown in Fig. 3 (a), the spatial patterns illustrated in Fig. 3 (b), and the recent system–task distributions summarized in the table. I reveals a gradual evolution of AI applications in water environmental systems. Early research focused on enhancing predictive performance under complex nonlinear conditions, while recent studies increasingly investigate AI support for system operation, intelligent control, and multi-objective optimization.

This background analysis provides the context for the task-oriented review presented in the subsequent sections, where AI

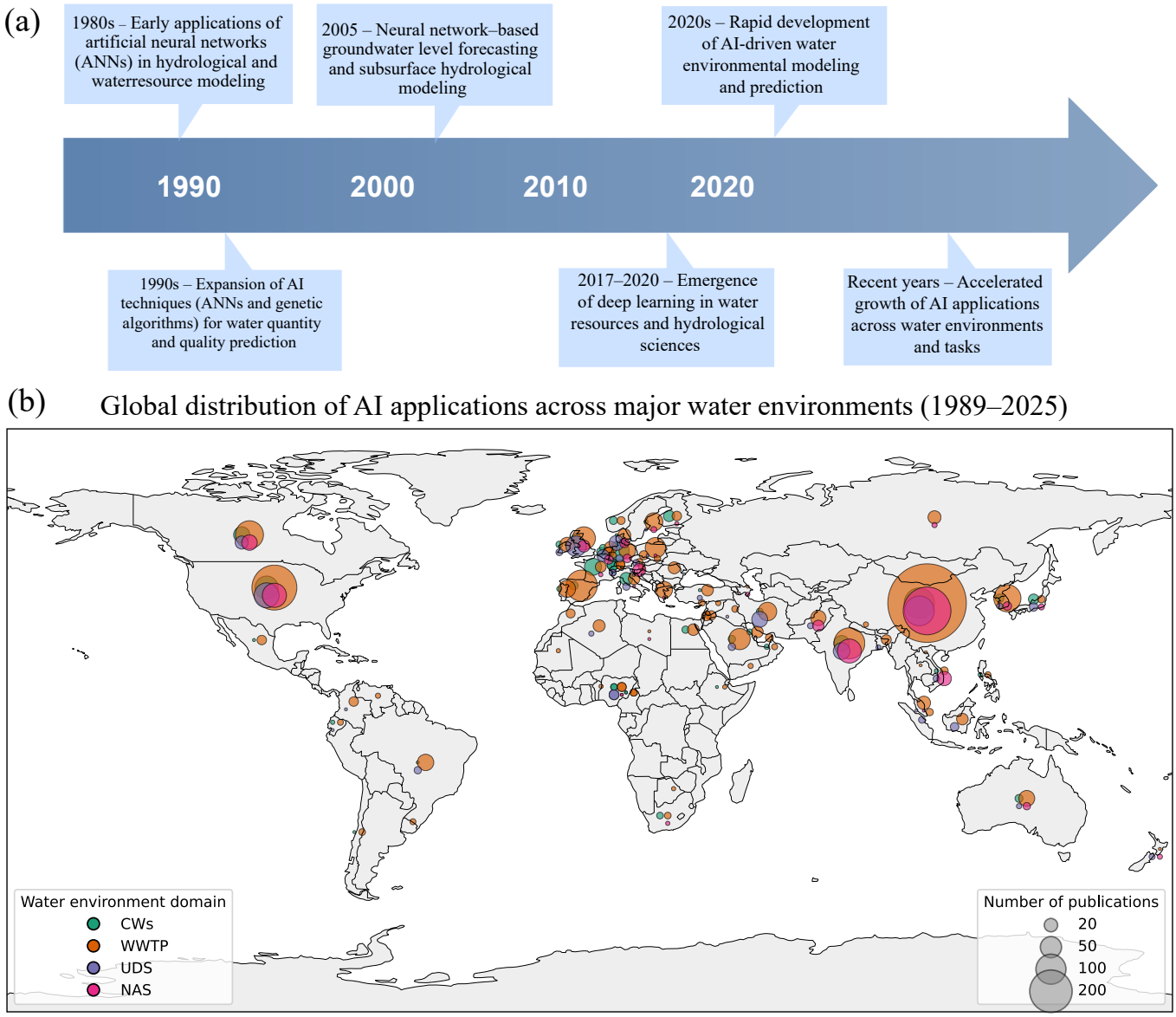

Fig. 3. **Historical evolution and global distribution of AI applications in water environmental research.** (a) Conceptual timeline of the methodological evolution of AI in water environmental research. (b) Global distribution of AI-related publications across major water environment domains—constructed wetlands (CWs), wastewater treatment plants (WWTPs), urban drainage systems (UDS), and natural aquatic systems (NAS)—from 1989 to 2025, where marker size indicates publication volume.

methods and applications are systematically examined across predictive modeling, pollution identification, intelligent control, multi-objective optimization, and knowledge discovery in different water environmental systems.

III. AI-Enabled Water Environmental Systems

Water environmental systems are not a single type of engineered entity, but rather a collection of complex systems composed of artificial infrastructures with varying degrees of engineering intervention and natural ecological processes. These systems include urban drainage systems, wastewater treatment plants, natural water bodies such as rivers, lakes, and reservoirs, as well as constructed wetlands. Fundamental differences exist among these systems in terms of operational objectives, process mechanisms, data structures, and controllability. Such differences lead to substantial variation in the depth, mode, and practical feasibility of AI integration.

Existing studies indicate that in highly engineered systems with clearly defined operational goals and stable monitoring and actuation interfaces, AI has gradually evolved from an auxiliary tool for offline modeling and predictive analysis into a core component supporting operational optimization and real-time control [111], [112]. In contrast, in systems dominated by ecological processes and natural evolution, AI is more commonly applied to state perception, process understanding, and environmental assessment [113]. This contrast highlights a pronounced imbalance in the maturity and functional roles of AI applications across different system types.

From a systems engineering perspective, such an imbalance

TABLE I
AI-RELATED STUDIES ACROSS WATER ENVIRONMENTAL SYSTEMS AND RESEARCH TASKS (2021–2025). NUMBERS INDICATE THE COUNT OF PUBLICATIONS (PERCENTAGE OF THE TOTAL IN PARENTHESES).

Water Environmental Systems	Predictive Modeling IV-A	Pollution Identification IV-B	Intelligent Control IV-C	Multi-objective Optimization IV-D	Knowledge Discovery IV-E
Urban Drainage Systems (UDS) III-B	7 papers: [12] [13] [14] [15] [16] [17] [18]	No paper in this area.	5 papers: [19] [20] [21] [22] [23]	4 papers: [24] [25] [26] [27]	No paper in this area.
Wastewater Treatment Plants (WWTP) III-C	19 papers: [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40] [41] [42] [43] [44] [45] [46]	8 papers: [47] [48] [49] [50] [51] [52] [53] [54]	9 papers: [55] [56] [57] [58] [59] [60] [61] [62] [63]	10 papers: [64] [65] [66] [67] [59] [68] [69] [70] [71] [72]	9 papers: [73] [74] [75] [76] [77] [78] [79] [80] [81]
Natural Aquatic Systems (NAS) III-D	8 papers: [82] [83] [84] [85] [86] [87] [88] [89]	No paper in this area.	No paper in this area.	No paper in this area.	1 paper: [90]
Constructed Wetlands (CWs) III-E	14 papers: [91] [92] [93] [94] [95] [96] [97] [98] [99] [100] [101] [102] [103] [104]	No paper in this area.	2 papers: [105] [106]	No paper in this area.	No paper in this area.

suggests that the applicability of AI is closely linked to intrinsic system characteristics, including the degree of engineering intervention, process formalization, and operational structure. However, how these system-level attributes constrain or enable different AI paradigms remains insufficiently articulated in existing reviews.

Based on this perspective, it is necessary to distinguish water environmental systems at the system level before discussing specific AI methods and tasks. This chapter, therefore, systematically reviews the engineering characteristics and current AI application status of four representative water environmental systems—urban drainage systems, wastewater treatment plants, natural aquatic systems, and constructed wetlands, providing a system-level contextual framework for the subsequent task-oriented analysis.

A. Coupling of Engineering Control and AI

Engineering controllability as a prerequisite for closed-loop AI. The uneven distribution of AI applications across water environmental systems primarily originates from differences in engineering controllability rather than algorithmic capability or data volume alone. In highly engineered systems such as wastewater treatment plants, AI operates within a closed decision–execution–feedback loop, where model outputs can be directly translated into actionable control strategies and evaluated through measurable system responses. This structural closure enables AI to move beyond descriptive modeling toward operational optimization and real-time control.

Breakdown of learning–action coupling in weakly controlled systems. In contrast, natural aquatic systems and constructed wetlands lack sufficient controllable degrees of freedom to support such closed-loop learning. Although AI can achieve high accuracy in state estimation or trend prediction, the absence of fine-grained intervention mechanisms prevents effective coupling between model outputs and system evolution. As a result, learning objectives remain weakly constrained, feedback signals are indirect or delayed, and AI applications are largely confined to offline analysis and environmental assessment.

Lack of system formalization and its impact on AI generalization. Furthermore, this imbalance is reinforced by fundamental differences in system formalization. Engineered systems benefit from standardized process representations, control variables, and performance metrics, which facilitate data integration, model transferability, and comparative evaluation across sites. By contrast, ecological and semi-natural systems are characterized by site-specific process descriptions and heterogeneous monitoring protocols, leading to fragmented data semantics and limited model generalization.

B. Urban Drainage Systems (UDS)

Urban drainage systems are typical event-driven infrastructures whose operational states are highly dependent on rainfall processes, exhibiting strong temporal variability and abrupt dynamics. Over short time scales, these systems may experience rapid and significant fluctuations in hydraulic and water quality conditions, rendering traditional steady-state or quasi-steady-state modeling approaches inadequate [114].

In terms of data characteristics, urban drainage systems often possess high temporal resolution monitoring data; however, such data are strongly non-stationary and uncertain [115]. Previous studies have shown that even in regions with relatively comprehensive sensor coverage, data loss and noise under extreme rainfall events are difficult to avoid [114].

From a control perspective, urban drainage systems exhibit relatively strong but highly constrained controllability, for example, through gates, pumping stations, and storage

facilities that regulate flow. Nevertheless, control actions must be executed within very short time windows to mitigate urban flooding risks and pollutant overflows. Consequently, AI models applied to this system are required to prioritize rapid inference, robustness, and operational safety rather than long-term predictive accuracy. These characteristics make urban drainage systems an important application domain for AI in emergency decision-making and real-time scheduling.

C. Wastewater Treatment Plants (WWTP)

Wastewater treatment plants (WWTP) are highly engineered, automated, and instrumented water environmental systems, with core objectives centered on stable and continuous operation under strict regulatory constraints. Modern WWTP are typically equipped with dense networks of online sensors and automated control units, enabling real-time acquisition of key operational variables such as flow rate, dissolved oxygen, nutrient concentrations, and energy consumption. These characteristics result in structured, continuous, and high-reliability data streams, making WWTP particularly well-suited for data-driven and AI-based methodologies.

More importantly, WWTP possess explicit and actionable control interfaces, including aeration intensity adjustment, sludge recirculation ratio control, and chemical dosing strategies. Coupled with clearly defined operational objectives—such as effluent quality compliance, energy minimization, and operational stability—this strong controllability has made WWTP one of the earliest and most extensively adopted water environmental systems for AI applications [111], [112]. Beyond analytical modeling, AI techniques in WWTP can be directly embedded into operational decision-making and control loops, bridging the gap between prediction and action.

Fig. 4 illustrates the integration of AI research tasks across different stages of the wastewater treatment process, highlighting how distinct AI tasks are associated with specific functional units. Predictive modeling is predominantly applied throughout the treatment train—from influent characterization to downstream effluent quality prediction, supporting short- and long-term forecasting of water quality indicators and process states. Intelligent control and multi-objective optimization are primarily concentrated in biologically intensive units, such as anaerobic, anoxic, and aerobic tanks, where dynamic process regulation is critical for balancing treatment performance, energy consumption, and operational stability.

In addition, pollution identification and knowledge discovery tasks are mainly associated with upstream influent analysis and overall system-level assessment. These tasks aim to characterize pollutant sources, identify emerging contaminants (e.g., microplastics), and extract latent patterns from long-term operational data. As shown in Fig. 4, WWTP also serve as a key interface between wastewater-borne microplastics and their potential release into aerosols, further expanding the scope of AI applications from conventional process optimization to environmental risk assessment.

As a result, AI in WWTP has progressively evolved from a supporting analytical tool into an operational component, with applications spanning system diagnosis, fault detection, intelligent control, and multi-objective optimization. This development trajectory has positioned WWTP as an important experimental platform for advanced AI paradigms, including reinforcement learning, multi-objective evolutionary algorithms, and digital twin technologies, reinforcing its central role in AI-driven water environmental research.

D. Natural Aquatic Systems (NAS)

Natural aquatic systems, including rivers, lakes, and reservoirs, are primarily governed by natural hydrological cycles and ecological mechanisms, with very limited direct human controllability. Compared with engineered systems, these systems emphasize long-term evolutionary behavior rather than immediate operational responses.

In terms of data sources, monitoring of natural aquatic systems typically relies on remote sensing observations, cross-sectional surveys, and a limited number of in situ sensors, resulting in multi-scale, heterogeneous datasets with inconsistent spatial and temporal resolutions. A substantial body of research shows that these data characteristics significantly challenge both process-based and data-driven models, particularly in scale coupling and uncertainty propagation [116].

Within this context, AI applications in natural aquatic systems are mainly oriented toward environmental sensing and system cognition, including spatial water quality mapping, abnormal event detection, and long-term trend analysis, rather than operational control [113]. This application orientation reflects the objective limitations on intervention in natural systems and positions AI as a supporting tool for environmental management and scientific understanding rather than direct decision execution.

E. Constructed Wetlands (CWs)

Constructed wetlands are representative semi-engineered ecological treatment systems, in which pollutant removal depends on the long-term coupling of hydraulic transport, microbial metabolism, plant uptake, and physicochemical processes. Compared with conventional water treatment facilities, constructed wetlands typically operate under passive or weakly controlled conditions, with limited operational variables beyond influent regulation and periodic maintenance. From a systems engineering perspective, this characteristic places constructed wetlands closer to ecological systems than to industrial processes.

At the monitoring level, data acquisition in constructed wetlands is generally characterized by low frequency, discreteness, and strong spatial heterogeneity. Common data forms include influent and effluent water quality measurements, intermittent sampling of substrates and vegetation, and limited hydraulic observations. Previous studies indicate that strong temporal and spatial non-uniformity of key biogeochemical processes in constructed wetlands hinders comprehensive state characterization under sparse sensor deployment.

Under these constraints, AI applications in constructed wetlands primarily function as auxiliary analytical tools rather than operational control mechanisms. Existing studies primarily address offline tasks such as performance prediction,

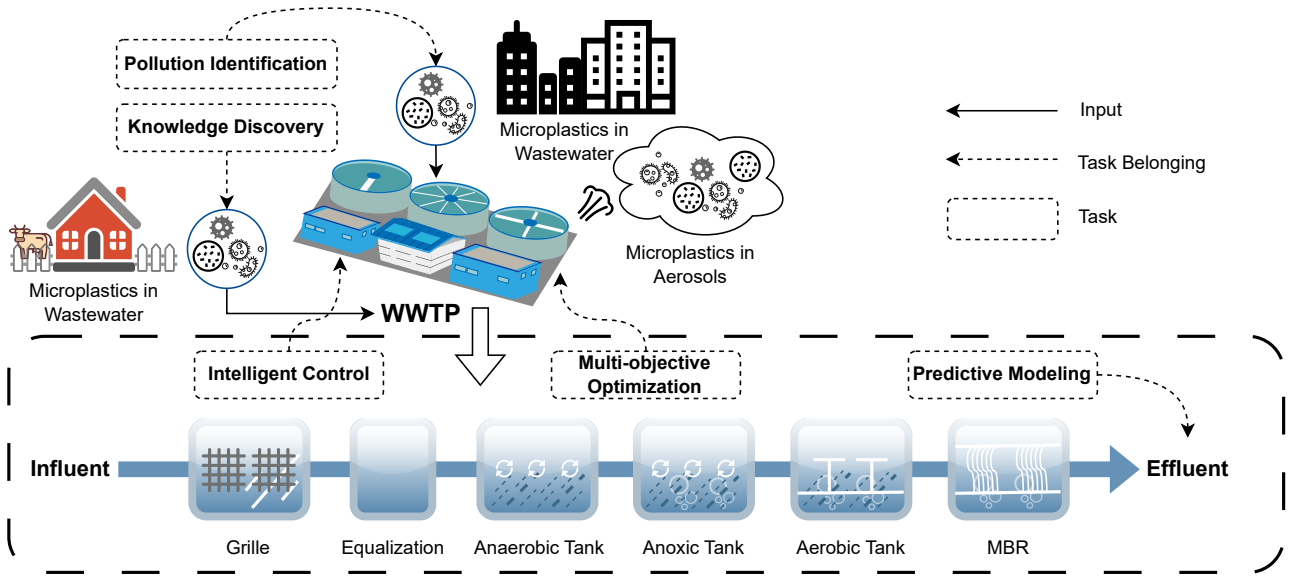

Fig. 4. **Schematic illustration of wastewater treatment processes and associated AI research tasks.** Dashed boxes indicate the five major AI tasks across water environmental systems—pollution identification, knowledge discovery, intelligent control, predictive modeling, and multi-objective optimization, and their corresponding positions within the wastewater treatment process.

scenario analysis, design optimization, and process understanding, often using machine learning to capture nonlinear relationships among hydraulic loading, temperature, vegetation configuration, and treatment efficiency. By contrast, real-time closed-loop control remains largely infeasible due to structural rather than algorithmic constraints.

IV. AI Methods for Water Environmental Tasks

This section provides a systematic overview of artificial intelligence (AI) methods used to address five core tasks in water environmental systems—predictive modeling, pollution identification, intelligent control, multi-objective optimization, and knowledge discovery. We examine how different AI paradigms have been adopted across urban drainage systems (UDS), wastewater treatment plants (WWTP), natural aquatic systems (NAS), and constructed wetlands (CWs).

For analytical clarity, the methods are grouped into three major paradigms: traditional machine learning (ML), deep learning (DL), and reinforcement learning (RL). Here, traditional ML refers to conventional algorithms such as regression models, support vector machines, and ensemble learning, explicitly excluding deep neural networks. Deep learning encompasses neural network–based architectures including convolutional, recurrent, transformer-based, and graph neural network models, while RL is distinguished by its state–action–reward decision-making mechanism. For each task, existing studies are reviewed under this tripartite framework, highlighting dominant approaches and underexplored directions.

Fig. 5 presents the methodological taxonomy of artificial intelligence approaches applied in water environmental research, organizing representative algorithms into three primary paradigms: traditional machine learning (ML), deep learning (DL), and reinforcement learning (RL). Table II further summarizes the distribution of these methods across major water environmental systems, including urban drainage systems (UDS), wastewater treatment plants (WWTP), natural aquatic systems (NAS), and constructed wetlands (CWs). The results indicate that traditional ML methods are widely applied across all system types, demonstrating broad adaptability. In contrast, advanced DL architectures and RL approaches are more concentrated in WWTP and UDS, with comparatively limited adoption in NAS and CWs.

A. Predictive Modeling

Predictive modeling represents the most mature and widely adopted AI task across water environmental systems, serving as the primary entry point for data-driven analysis in both engineered and natural contexts. Existing studies aim to approximate complex, nonlinear relationships between system states and environmental or operational drivers, under conditions characterized by uncertainty, nonstationarity, and heterogeneous data availability. As summarized in Table II, a broad spectrum of AI methods has been employed for predictive purposes, ranging from traditional machine learning models to deep learning architectures and, more recently, reinforcement learning frameworks.

1) Traditional Machine Learning (ML): Traditional machine learning methods—excluding deep neural networks—are widely used for predictive modeling across all four water environmental systems due to their low data demands, computational efficiency, and interpretability. Linear regression (LR) [91], [103], support vector regression (SVR) [16], [94], [97], [101], random forests (RF) [18], [29], [32], [93], XGBoost [33], [89], [95], and Bayesian [17] models are commonly applied to predict water quality, pollutant levels, hydraulic responses, and treatment performance.

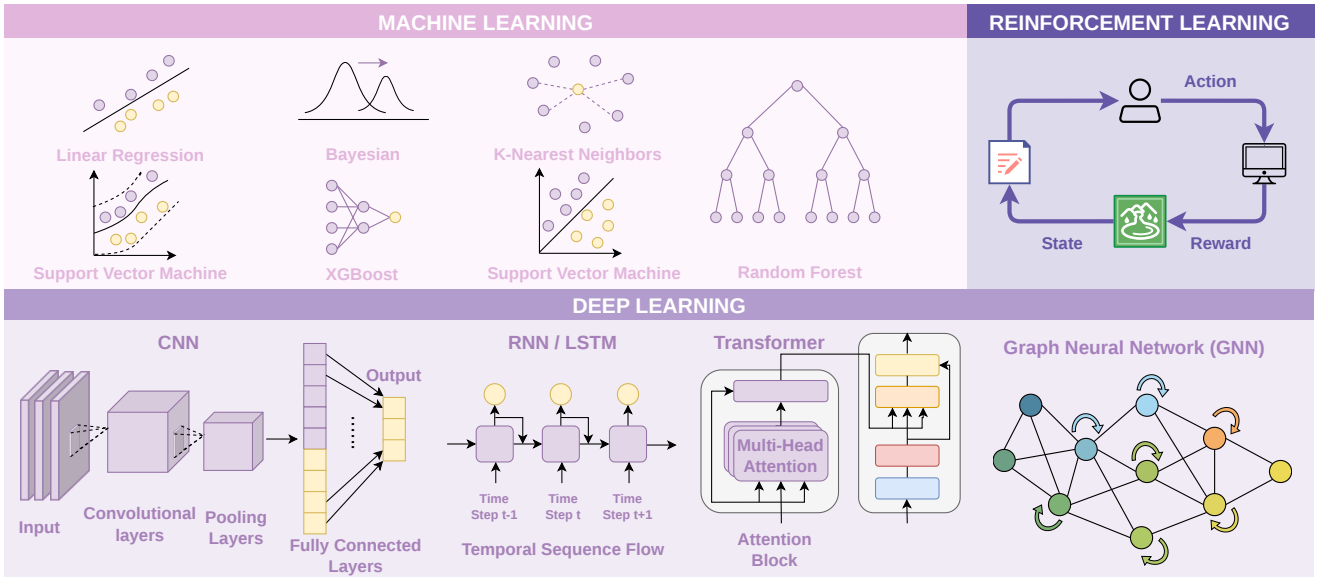

Fig. 5. **Taxonomy of artificial intelligence (AI) methods applied in water environmental systems.** The figure presents a hierarchical categorization of AI approaches into three primary paradigms: **machine learning (ML), deep learning (DL), and reinforcement learning (RL)**. Representative algorithms are illustrated within each category. Machine learning includes classical methods such as linear regression, Bayesian models, k-nearest neighbors, support vector machines, XGBoost, and random forests. Deep learning covers convolutional neural networks (CNN), recurrent neural networks (RNN/LSTM), Transformer architectures, and graph neural networks (GNN). Reinforcement learning is shown separately to emphasize its closed-loop decision-making mechanism based on state–action–reward interactions.

In engineered systems such as wastewater treatment plants and urban drainage systems, tree-based ensemble models (RF and XGBoost) are widely adopted due to their strong robustness to complex nonlinear dynamics, noisy measurements, and partially missing data. In wastewater treatment plants, these models are frequently used for effluent quality and overall process performance prediction, while Bayesian approaches are further employed to explicitly quantify uncertainty under highly fluctuating and variable influent conditions. In urban drainage systems, traditional ML models are particularly effective for short-term rainfall–runoff and flood-related forecasting tasks typically characterized by strong nonstationarity and very high temporal resolution.

In natural aquatic systems and constructed wetlands, traditional ML models are primarily applied to predictive tasks under relatively sparse and heterogeneous real-world data conditions. RF and SVR are commonly used to estimate nutrient concentrations and key water quality indicators from limited in situ and remote sensing observations, while in constructed wetlands, these methods are employed to predict pollutant removal efficiency and overall effluent quality under varying hydraulic loading and environmental operating conditions [91], [100], [101]. Despite their broad practical applicability in diverse scenarios, traditional ML approaches generally rely on manual feature engineering and exhibit limited capability in modeling long-term temporal dependencies.

2) Deep Learning (DL): Deep learning models have become the dominant paradigm for predictive modeling in recent studies, particularly in data-rich engineered systems. Feed-forward neural networks and convolutional neural networks (CNN) [12], [13], [99] are commonly used to model nonlinear relationships between multivariate inputs and system outputs, while recurrent architectures, including RNN, LSTM [28], [30], [30], [36], and GRU [39], [98], are specifically designed to capture temporal dependencies in sequential water environment data.

In wastewater treatment plants, LSTM-based models are extensively applied to predict effluent water quality, nutrient dynamics, and internal process states under dynamic influent conditions. Their ability to represent delayed system responses and memory effects enables improved performance over traditional ML methods, especially for multistep forecasting tasks. More recently, Transformer-based architectures and attention mechanisms have been introduced to enhance the modeling of long-range temporal dependencies and complex variable interactions, although their application remains largely limited to systems with dense and continuous monitoring data.

Neural network models are also widely used in urban drainage systems for short-term flood and flow prediction, where rapid system response and strong temporal variability dominate system behavior. In contrast, their adoption in natural aquatic systems and constructed wetlands remains more constrained by data availability and sampling heterogeneity. While LSTM-based models have been explored for water quality prediction in rivers and lakes, most studies in constructed wetlands focus on performance prediction and scenario analysis using relatively shallow architectures [98], [99], [103]. Advanced architectures such as GRUs, Transformers, and GNNs are rarely reported in CW-related predictive studies, reflecting both data limitations and the weak controllability of these systems.

3) Reinforcement Learning (RL): Reinforcement learning is fundamentally designed for sequential decision-making rather than static input–output prediction. Unlike supervised ML and

DL models that directly learn mappings between historical observations and target variables, RL optimizes action policies through state–action–reward interactions within an environment. As such, it does not naturally align with standalone predictive modeling tasks.

To date, RL has rarely been employed as an independent predictive tool in practical water environmental systems. This is not primarily due to data scarcity or inherent technical infeasibility, but rather because most prediction problems fundamentally lack the interactive structure required for proper RL formulation. In typical predictive settings, there is no explicit action space, no iterative feedback loop, and no well-defined reward function to effectively guide policy learning. Introducing RL into such contexts would therefore require artificial construction of decision-making environments, which offer little clear methodological advantage over conventional supervised learning approaches.

Consequently, RL applications in water environmental research are almost exclusively embedded within control or optimization frameworks, where sequential decisions and reward signals are inherently defined. Its absence in standalone predictive modeling reflects a structural mismatch between RL's decision-oriented learning paradigm and the static nature of prediction tasks.

4) Practical Deployment and Operational Reliability: Despite the large volume of predictive modeling studies reporting high accuracy under experimental settings, only a limited subset of these models has been demonstrated to support real-time operational decision making. Most predictive models are developed and evaluated using historical datasets under controlled assumptions, whereas real-world deployment requires sustained performance under sensor failures, data drift, unobserved disturbances, and changing operational regimes.

A key gap lies in the mismatch between accuracy-oriented evaluation metrics and operational reliability requirements. While metrics such as RMSE or R^2 effectively quantify average predictive performance, they provide limited insight into robustness under extreme events, distribution shifts, or rare but critical conditions. As a result, models with superior benchmark accuracy may exhibit unstable behavior when embedded into real-time scheduling or control systems.

Consequently, industrial adoption of predictive AI models remains selective, favoring approaches that trade marginal accuracy gains for improved robustness, interpretability, and fail-safe behavior. This highlights the need to complement prediction accuracy with deployment-oriented evaluation criteria, including uncertainty awareness, stability under missing data, and graceful degradation under abnormal conditions.

B. Pollution Identification

Pollution identification focuses on detecting, classifying, and attributing pollution events, contamination sources, and abnormal water quality patterns in complex environmental systems. Unlike predictive modeling, which estimates future system states, pollution identification emphasizes event recognition, anomaly detection related to contamination, and source attribution under heterogeneous monitoring conditions. AI methods for this task are therefore evaluated primarily by detection reliability, timeliness, and robustness in supporting environmental risk assessment and early warning.

1) Traditional Machine Learning (ML): Traditional machine learning methods—excluding deep neural networks—have been widely adopted for pollution identification due to their interpretability and stability under limited data availability. Typical applications include contamination classification, pollution source attribution, and anomaly detection in physicochemical monitoring data.

In constructed wetlands, traditional ML models are applied to identify abnormal pollutant removal behavior and detect potential contamination events under fluctuating environmental conditions [113]. In natural aquatic systems, models such as random forests [47], [54] and support vector machines [51] are commonly used to classify pollution sources and distinguish anthropogenic contamination from natural variability. These approaches are particularly effective in data-scarce environments where model transparency and stability are critical for regulatory interpretation.

2) Deep Learning (DL): Deep learning models have increasingly been employed for pollution identification, particularly when handling high-dimensional, multi-source, or spatiotemporal data. Convolutional neural networks (CNN) [55], recurrent neural networks (RNN), and related architectures are commonly used for anomaly detection, contamination classification, and pollution event recognition in monitoring time series and remote sensing imagery.

In wastewater treatment plants, neural networks are increasingly applied to detect abnormal operational states associated with potential effluent violations or sudden contaminant shocks. Their strong capacity to learn complex nonlinear interactions among biological, chemical, and hydraulic process variables enables earlier and more reliable detection of contamination-related disturbances compared to conventional threshold-based monitoring methods.

In large-scale natural water bodies, representation learning methods have also been explored for large-area contamination monitoring using remote sensing data [113]. These studies demonstrate the effectiveness of DL in identifying complex and spatially distributed pollution patterns, although interpretability remains an ongoing challenge.

3) Reinforcement Learning (RL): Reinforcement learning has not been directly applied as a primary methodology for pollution identification in water environmental systems. Pollution identification tasks are typically formulated as classification, source apportionment, or anomaly detection problems, which rely on learning mappings between observed features and predefined pollution categories or source labels. These problem structures naturally align with supervised or unsupervised learning paradigms rather than sequential decision-making frameworks.

While RL has been explored in related contexts such as monitoring strategy optimization or sensor placement planning, these applications belong to adaptive resource allocation or control tasks rather than pollution identification itself. Therefore, the absence of RL in standalone pollution identification reflects a structural mismatch between RL's decision-

TABLE II
SUMMARY OF AI METHODS APPLIED ACROSS DIFFERENT WATER ENVIRONMENTAL SYSTEMS.

Category	Method	Water Environmental Systems			
		UDS	WWTP	NAS	CWs
Traditional ML	LR	✓	✓	✓	✓
	Bayesian	✓	✓	✗	✗
	KNN	✗	✗	✗	✗
	SVM	✓	✓	✓	✓
	SVR	✓	✓	✓	✓
	RF	✓	✓	✓	✓
	XGBoost	✓	✓	✓	✓
Deep Learning	RNN	✓	✓	✓	✗
	LSTM	✓	✓	✓	✗
	CNN	✓	✓	✓	✗
	GNN	✗	✓	✓	✗
	GRU	✗	✓	✗	✗
	Transformer	✗	✓	✗	✗
Reinforcement Learning	RL	✓	✓	✗	✗

oriented learning paradigm and the static inference nature of identification problems, rather than a temporary research gap.

4) Practical Deployment and Decision Relevance: AI-based pollution identification methods have achieved notable and well-documented success in detecting anomalies, emerging contaminants, and diverse pollution sources under controlled experimental or retrospective analytical settings. However, their translation into real-world operational environmental management remains constrained not by classification accuracy alone, but by uncertainty propagation, overall system reliability, and practical decision relevance.

In practical settings, pollution identification systems must support regulatory response, risk assessment, and early warning mechanisms, where false positives, missed detections, and delayed alerts can incur substantial operational and social costs. Many studies emphasize accuracy metrics under controlled datasets, yet provide limited evaluation of detection latency, robustness under sensor noise, cross-site generalization, or resilience to distribution shifts.

Consequently, real-world deployment tends to favor interpretable, conservative, and multi-source corroborated identification frameworks over highly complex models with marginal performance gains. Future research should therefore prioritize uncertainty quantification, robustness assessment, and integration with operational decision workflows to enhance practical reliability and trustworthiness.

C. Intelligent Control

Intelligent control represents a critical transition of AI applications in water environmental systems from passive monitoring and prediction toward active decision-making and real-time operation. Unlike predictive modeling, where AI primarily serves as a surrogate for system dynamics, intelligent control focuses on embedding data-driven models directly into control loops to optimize operational objectives under dynamic and uncertain conditions. Existing studies can be broadly categorized into traditional machine learning –assisted control, deep learning –enhanced control, and reinforcement learning –driven control frameworks.

1) Traditional Machine Learning (ML): Traditional machine learning methods—excluding deep neural networks—are commonly employed as auxiliary components within model-based or rule-based control frameworks. In this paradigm, ML models typically function as soft sensors, surrogate process models, or decision-support tools, while the core control logic remains externally defined.

In constructed wetlands, early studies emphasized adaptive and information-driven control concepts rather than fully autonomous control. Kamal [105] highlighted the potential of artificial intelligence to enhance operational adaptability by learning nonlinear relationships among influent conditions, environmental factors, and treatment performance. Related work by Li *et al.* [106] integrated machine learning with online monitoring to support operational decision-making in wetland landscape systems, illustrating the use of ML as an assistive rather than directive control layer.

In wastewater treatment plants, ML-assisted control has been more systematically explored. Wang *et al.* [61], [117] demonstrated how supervised learning and data-driven feature engineering can support intelligent aeration control, while Lai [57] incorporated effluent quality prediction into an integrated control framework without replacing conventional controllers. Dai *et al.* [64] further extended this concept to coupled WWTP–constructed wetland systems, reinforcing the role of traditional ML as a supervisory optimization layer.

2) Deep Learning (DL): Deep learning models provide more expressive representations of nonlinear process dynamics and are increasingly embedded deeper into control architectures, reducing reliance on explicit mechanistic modeling compared with traditional ML-assisted approaches.

In wastewater treatment plants, neural networks have been used to directly support or inform control decisions. Ankalaki [58] applied neural architecture search to automate wastewater characteristic parameter quantification, enabling adaptive control based on learned spectral representations. Prabu *et al.* [62] proposed aquaflownet, a neural network-based framework for real-time wastewater flow management, where learned models facilitated operational optimization under dynamic conditions.

Hybrid supervisory control strategies combining neural networks with optimization have also been explored. Heo *et al.* [68] developed a multi-objective supervisory control framework that integrates neural network predictions with optimization-based decision making to balance treatment efficiency, energy consumption, and operational cost at full scale.

Although graph neural networks (GNNs) have demonstrated strong capability in modeling spatially structured processes, their direct application as standalone controllers remains limited. Existing graph-based approaches are typically integrated within broader deep learning or multi-agent reinforcement learning frameworks rather than functioning as independent GNN controllers.

Despite their improved flexibility under nonlinear conditions, most DL-based control approaches remain embedded within supervisory or hybrid frameworks rather than fully end-to-end control systems, reflecting practical concerns regarding stability, interpretability, and robustness.

3) Reinforcement Learning (RL): Reinforcement learning, particularly deep reinforcement learning, has emerged as

the dominant paradigm for intelligent control in water environmental systems by formulating control as a sequential decision-making problem and learning optimal policies through environment interaction.

In wastewater treatment plants, DRL has been extensively applied to aeration, pumping, and multi-objective process control. Seo *et al.* [63] demonstrated energy-efficient joint pumping control using DRL under dynamic influent conditions, while Aponte-Rengifo *et al.* [60] proposed a model-free DRL framework that bypasses explicit process modeling. Mohammadi *et al.* [56] further enhanced robustness by applying soft actor–critic algorithms with time-delay considerations. Unified and multi-objective DRL control strategies have also been explored, exemplified by the coordinated control framework proposed by Croll *et al.* [59].

Urban drainage systems constitute another major application domain of DRL-based control. Tian *et al.* [19], [22] combined DRL with model predictive control to mitigate flooding and combined sewer overflows, with subsequent studies addressing real-time performance [20], knowledge–data fusion for training acceleration [21], and LSTM-enhanced policy learning [23]. Recent extensions include pollution-aware integrated control [25] and multi-agent DRL frameworks for large-scale drainage networks [26], [118].

Compared with ML- and DL-assisted supervisory strategies, RL-based control frameworks shift the control logic itself into a learned policy, enabling adaptive and closed-loop optimization under complex and uncertain operational conditions.

4) Deployment Readiness and Control Reliability: Although intelligent control represents the most direct pathway for AI to influence system operation, the gap between experimental performance and practical deployment remains substantial. Many control-oriented studies demonstrate improvements in simulation or pilot-scale environments, yet full-scale adoption is often limited by concerns regarding stability, safety, and reliability under unanticipated conditions.

In operational settings, control systems must satisfy strict requirements on robustness, constraint handling, and fail-safe behavior. Data-driven controllers, particularly those based on deep reinforcement learning, may exhibit sensitivity to modeling errors, delayed feedback, or distributional shifts, which undermines operator trust despite favorable average performance. Consequently, most deployed AI control solutions remain embedded within supervisory or hybrid frameworks rather than operating as fully autonomous controllers.

This disparity highlights that control performance alone is insufficient for real-world adoption. Deployment readiness depends on demonstrable robustness, transparency of decision logic, and compatibility with existing control infrastructures, explaining why industrial implementations remain selective despite rapid methodological advances.

D. Multi-objective Optimization

Multi-objective optimization addresses one of the most fundamental challenges in water environmental systems: the simultaneous satisfaction of competing operational goals, such as effluent quality, energy consumption, economic cost, system stability, and environmental risk. Unlike single-objective prediction or control tasks, multi-objective optimization explicitly models trade-offs and seeks Pareto-optimal solutions under complex constraints. AI-based approaches for this task can be broadly grouped into traditional machine learning–assisted optimization, deep learning–based surrogate optimization, and reinforcement learning–driven multi-objective optimization.

1) Traditional Machine Learning (ML): Traditional machine learning methods—excluding deep neural networks—are commonly used as surrogate models or auxiliary tools within classical multi-objective optimization frameworks. In this setting, ML models approximate computationally expensive process simulations or infer relationships among decision variables and objectives, while the optimization itself is typically performed using evolutionary algorithms or mathematical programming techniques.

In wastewater treatment plants, early multi-objective optimization studies frequently combined traditional ML with heuristic optimization algorithms. Dai *et al.* [64] integrated machine learning-based decision support into a combined WWTP–constructed wetland system, optimizing treatment efficiency and resource utilization.

In urban drainage systems, traditional ML-assisted optimization has been applied primarily to system design and operational planning rather than real-time control. Wang *et al.* [67] employed data-driven models to support multi-objective optimization of drainage performance, balancing flood mitigation and infrastructure cost. These studies highlight the role of traditional ML as an enabling approximation layer that reduces computational burden and enhances optimization feasibility, rather than directly performing multi-objective search.

2) Deep Learning (DL): Deep learning models provide expressive nonlinear surrogates that are particularly suitable for modeling complex, high-dimensional objective spaces in multi-objective optimization problems. Compared with traditional ML models, neural networks capture stronger nonlinear interactions among variables and competing objectives.

In WWTPs, neural network-based surrogate models have been widely used to support multi-objective optimization of process operation. Heo *et al.* [68] proposed a hybrid machine learning-based multi-objective supervisory control strategy, where neural networks were trained to predict multiple performance indicators and guide optimization under conflicting objectives. Such approaches enable efficient approximation of Pareto fronts without repeatedly invoking computationally intensive mechanistic simulations.

Neural network-based optimization has also been explored in broader water environmental systems. Prabu *et al.* [62] demonstrated how neural network-based flow modeling can support multi-objective decision making in wastewater flow management, implicitly balancing hydraulic efficiency and treatment performance. In addition, graph neural networks (GNNs), as a branch of deep learning, have begun to support topology-aware surrogate modeling in spatially interconnected systems, although their explicit application in standalone multi-objective optimization remains limited.

Overall, DL primarily functions as a high-fidelity objective approximator within optimization pipelines, enhancing scal-

ability and representation capacity under complex nonlinear conditions.

3) Reinforcement Learning (RL): Reinforcement learning, particularly deep reinforcement learning (DRL), has become increasingly prominent in multi-objective optimization due to its ability to directly learn decision policies under competing objectives. In contrast to surrogate-based optimization, RL integrates optimization and control into a unified sequential decision-making framework, where multiple objectives are encoded within reward structures or policy constraints.

In WWTPs, several studies have formulated operational optimization as a multi-objective RL problem. Croll *et al.* [59] proposed a unified deep reinforcement learning framework that jointly optimizes multiple operational objectives, demonstrating improved performance over single-objective control strategies. Mohammadi *et al.* [56] further incorporated time-delay effects into multi-objective RL-based optimization, enhancing robustness under realistic operational constraints.

Urban drainage systems represent a particularly active application domain for RL-based multi-objective optimization. Tian *et al.* [22] combined deep reinforcement learning with model predictive control to simultaneously optimize flood mitigation and overflow reduction. Subsequent studies extended this framework to incorporate additional objectives such as pollution control [25], operational reliability [20], and long-term system resilience [19]. Multi-agent reinforcement learning has also been explored to coordinate distributed optimization across large-scale drainage networks [26], [118].

Graph-based representations are often embedded within RL frameworks to encode network topology and spatial dependencies, particularly in multi-agent settings. Rather than serving as independent optimization methods, graph learning components typically enhance state representation within RL-based multi-objective decision-making architectures.

4) Operational Feasibility and Solution Robustness: AI-based multi-objective optimization methods have shown strong potential in identifying Pareto-optimal trade-offs among competing objectives under modeled conditions. However, their practical adoption is often constrained by the sensitivity of optimized solutions to modeling assumptions, data uncertainty, and unmodeled operational constraints.

In real-world applications, operators tend to favor solutions that are robust across a wide range of operating scenarios rather than those optimized for nominal conditions. Many optimization studies report well-defined Pareto fronts without sufficiently analyzing solution stability under uncertainty or long-term operational variability, limiting their direct applicability in decision support systems.

As a result, industrial use of AI-based multi-objective optimization often emphasizes conservative operating envelopes and scenario-robust solutions, even at the expense of theoretical optimality. Bridging this gap requires shifting evaluation focus from Pareto optimality alone toward robustness, constraint satisfaction, and operational resilience.

E. Knowledge Discovery

Knowledge discovery focuses on extracting interpretable patterns, latent structures, and governing relationships from complex water environmental data to advance system understanding rather than directly support operational decision making. In contrast to predictive modeling or intelligent control, this task emphasizes explanation, hypothesis generation, and mechanism-oriented insight. AI methods for knowledge discovery are therefore primarily evaluated by their ability to reveal meaningful system behavior under heterogeneous and uncertain conditions, rather than by predictive accuracy or control performance alone.

1) Traditional Machine Learning (ML): Traditional machine learning methods—excluding deep neural networks—have been widely adopted for knowledge discovery due to their relative interpretability and robustness under limited data availability. Common applications include feature importance analysis, sensitivity assessment, clustering, rule extraction, and causal inference support.

In constructed wetlands, traditional ML has been extensively used to explore nonlinear relationships between environmental factors and treatment performance. Chen *et al.* [71] applied machine learning models to identify key drivers of pollutant removal efficiency, revealing the dominant influence of hydraulic loading and temperature under varying operational conditions. Similar approaches were reported by Almegdadi *et al.* [113], where data-driven analysis supported process understanding rather than direct optimization.

In natural aquatic systems, traditional ML methods such as random forests and support vector machines have been used to infer spatial and temporal drivers of water quality variability. Lowe *et al.* [116] highlighted that these models enable interpretable assessment of factor contributions, supporting environmental management and scientific analysis under data scarcity. Such studies demonstrate that traditional ML serves as an effective bridge between observational data and conceptual system understanding.

2) Deep Learning (DL): Deep learning models have increasingly been employed for knowledge discovery in water environmental systems, particularly when dealing with high-dimensional, multi-source, or spatiotemporal data. While neural networks are often criticized for limited interpretability, recent studies demonstrate their ability to extract latent representations that correspond to meaningful environmental processes when combined with post hoc analysis or representation learning techniques.

In wastewater treatment plants, neural networks have been applied to uncover hidden process dynamics and operational regimes. Yang *et al.* [112] showed that deep learning effectively captures complex bio-chemical-hydraulic interactions, providing deeper insights into process stability and system sensitivity beyond conventional mechanistic models.

In natural water bodies, neural networks have been widely used to analyze remote sensing and long-term monitoring data for pattern discovery. Almegdadi *et al.* [113] showed that self-supervised learning extracts intrinsic water quality representations from unlabeled data, enabling large-scale anomaly detection and system characterization.

Graph neural networks (GNNs), as a branch of deep learning, further extend knowledge discovery capabilities by explicitly encoding spatial connectivity and network topology.

In river networks and urban drainage systems, GNN-based models have facilitated the identification of spatial interaction patterns and propagation mechanisms that are difficult to capture using independent feature representations.

3) Reinforcement Learning (RL): Reinforcement learning plays a comparatively limited role in knowledge discovery, as it is inherently oriented toward sequential decision making and policy optimization. Unlike ML and DL models that directly analyze data structures, RL primarily learns action policies through interaction with environments.

However, trained RL policies can be systematically analyzed post hoc to infer underlying system sensitivities, trade-off preferences, and implicit control priorities. For example, studies such as Croll *et al.* [59] suggest that carefully examining learned value functions or policy response patterns can reveal which critical system states exert the strongest influence on overall decision strategies. In this sense, RL contributes indirectly to broader knowledge discovery by exposing the internal structure of learned control logic rather than serving as a primary standalone discovery methodology.

4) Practical Impact and Knowledge Utilization: For knowledge discovery tasks, the primary challenge is not immediate real-time deployment, but rather the effective and meaningful utilization of extracted insights in scientific analysis and practical environmental management. While AI-based approaches can reveal latent patterns and complex underlying relationships, their broader practical impact ultimately depends on interpretability, consistency across diverse datasets, and close alignment with established domain knowledge.

Highly expressive models often produce abstract representations that are difficult to translate into actionable insight or management guidance, limiting their practical use to exploratory analysis. In contrast, models with transparent factor attribution, stable cross-scenario behavior, or physically interpretable structures are more readily integrated into environmental assessment and policy workflows.

Therefore, the practical value of AI-based knowledge discovery lies less in model complexity and more in its ability to support explanation, hypothesis testing, cross-site comparison, and mechanism refinement. This reinforces the importance of interpretability and conceptual coherence over predictive performance alone in discovery-oriented applications.

V. Literature Overview and Annual Publication Trend Analysis

Fig. 3 presents an overview of the temporal evolution of AI applications in water environmental science together with their global publication distribution. Fig. 6 shows the annual publication trends across different water environment systems. From the annual variation, it is evident that research on wastewater treatment plants (WWTP) has experienced an explosive growth since 2020, indicating that wastewater treatment remains the core focus in AI applications for water environmental systems.

Research on urban drainage systems (UDS) and natural water bodies (NAS) shows a steady increase over the years, suggesting these systems are gradually gaining attention, although the scale of research remains smaller than that of

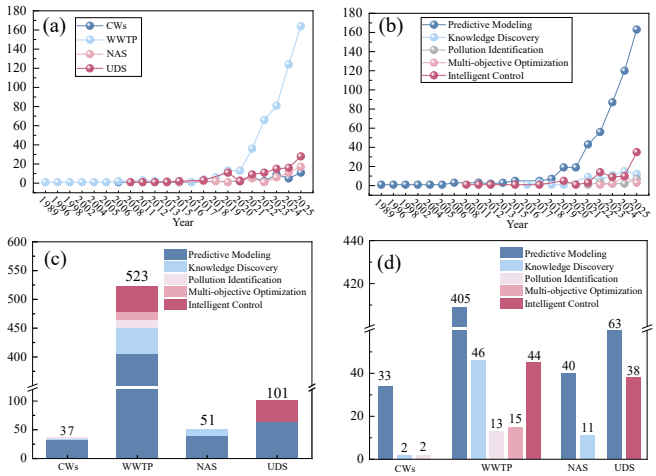

Fig. 6. **Publication trends and task distribution across water environmental systems.** (a) Annual publication trends in four water environmental systems, including constructed wetlands (CWs), wastewater treatment plants (WWTPs), natural aquatic systems (NAS), and urban drainage systems (UDS). (b) Temporal trends of the five major AI tasks across all systems. (c) Cumulative publication volume of different AI tasks in each water environmental system. (d) Task-wise publication distribution within each system.

WWTP. In contrast, constructed wetlands (CWs) have started relatively late, and the total number of publications is still limited, reflecting that AI applications in this system are still at an exploratory stage.

From Fig. 6, which presents the total publication volume per system, it can be seen that WWTP has the highest total publications (approximately 550). Overall, there is a significant imbalance in research efforts among different systems, likely influenced by differences in technological maturity, data accessibility, and practical application complexity.

A. Historical Development and International Distribution

Fig. 3 summarizes both the historical timeline of AI applications in water environmental science and the international distribution of publications.

The historical development can be outlined as follows: during the 1980s, artificial neural networks (ANNs) were first applied to hydrological and water resources modeling. In the 1990s, ANN techniques, often combined with genetic algorithms, were extended to water quantity and quality prediction. Around 2005, neural networks were applied to groundwater level prediction and hydrological modeling. Between 2017 and 2020, deep learning began to influence water resources and water environmental research. In recent years, AI applications have accelerated across multiple systems, including WWTP, UDS, NAS, and CWs.

Internationally, publication output is concentrated in China, the United States, and several European countries. Different systems are represented unevenly across regions: WWTP and predictive tasks have the highest publication volume in China and the United States, whereas NAS and CWs research has a notable presence in Europe and Southeast Asia. This indicates a clear national and system concentration effect, with

research hotspots located in countries and systems where data availability and technical maturity are relatively high.

B. Trends in Research Tasks Distribution

Fig. 6 illustrates the annual publication trends across five major AI research tasks in water environmental systems. Among the tasks considered, predictive modeling shows the fastest growth and the highest total publication volume (approximately 600), particularly after 2020, indicating that prediction remains the most mature and widely studied AI task in the water environmental systems.

Intelligent control and knowledge discovery analysis have demonstrated noticeable growth in recent years, though their total publication volumes remain below that of predictive tasks. Meanwhile, multi-objective optimization and pollution identification have relatively fewer publications but exhibit steady upward trends, suggesting that these emerging tasks are gradually receiving more attention.

Overall, AI applications in water environmental systems exhibit a task-oriented and scenario-specific pattern: predictive modeling dominates, while control and pollution detection tasks progressively enrich the research landscape.

C. Comprehensive Analysis and Future Trends

From both system- and task-level perspectives, WWTP emerges as the core system in AI-enabled water environmental research, while other systems, though slower in growth, are gradually becoming more active. Predictive modeling remains the dominant task, whereas intelligent control, knowledge discovery analysis, multi-objective optimization, and pollution identification are steadily rising, indicating a trend toward multi-task integration in the field.

Historically, AI applications have evolved from ANN to neural networks and finally to deep learning, concentrated in regions with mature technology and accessible data. Overall, future research is likely to emphasize the integration of multiple systems, tasks, and objectives, aiming to achieve intelligent management and optimization of water environmental systems.

VI. Gaps Across Functional AI Tasks

Despite rapid progress in applying artificial intelligence to water environmental systems, clear gaps persist across functional AI tasks when examined from a task-oriented perspective rather than a system- or method-driven viewpoint. Existing studies tend to advance individual tasks in isolation, while cross-task integration, methodological consistency, and system-level generalization remain limited.

A. Imbalanced Task Development and Overconcentration

A prominent gap lies in the uneven development of functional AI tasks. Predictive modeling and intelligent control dominate current research efforts, driven by relatively mature datasets, well-defined objectives, and direct operational relevance. In contrast, tasks such as pollution identification and knowledge discovery remain underrepresented, particularly in complex or open environmental settings. This imbalance limits the holistic intelligence of AI-enabled water environmental systems, as accurate prediction or control alone cannot fully support environmental decision-making without robust perception and knowledge extraction capabilities.

Moreover, even within extensively studied tasks, research often focuses on narrow performance metrics, such as prediction accuracy or control efficiency, while neglecting uncertainty quantification, interpretability, and long-term system resilience. As a result, advances at the task level do not always translate into reliable or transferable system-level intelligence.

1) Future Directions: Future research should promote a more balanced development of functional AI tasks by encouraging cross-task benchmarking and interdisciplinary dataset construction. Dedicated efforts are needed to strengthen underrepresented tasks such as pollution identification and knowledge discovery, particularly in open and heterogeneous environments. Evaluation metrics should expand beyond accuracy or efficiency to include uncertainty quantification, interpretability, and long-term system resilience. Establishing standardized multi-task benchmark platforms may further incentivize comprehensive task development rather than performance optimization within isolated domains.

B. Fragmentation Across Tasks and Lack of Task Coupling

Another critical gap is the fragmentation of AI tasks across the water system lifecycle. Most studies design and evaluate AI models for a single functional task, assuming that upstream sensing, downstream decision-making, or contextual knowledge is externally provided. This decoupled paradigm overlooks strong interdependencies among tasks, such as how pollution identification informs control objectives or how knowledge discovery guides predictive modeling.

The absence of explicit task coupling leads to redundant modeling efforts, inconsistent data representations, and suboptimal decision pipelines. Current frameworks rarely support closed-loop learning across multiple tasks, where insights generated by one task dynamically improve others. Addressing this gap requires a shift from task-specific optimization toward task-aware and task-coordinated AI architectures.

1) Future Directions: Addressing task fragmentation requires the development of integrated multi-task learning frameworks and closed-loop AI architectures. Future systems should enable structured information flow across perception, prediction, control, and knowledge modules through shared latent representations or graph-based task dependency modeling. Interoperable task interfaces and unified data schemas will be essential for reducing redundancy and improving cross-task consistency. Evaluation protocols should also measure end-to-end decision quality rather than isolated task performance.

C. Limited Generalization and Cross-System Transferability

While AI methods have demonstrated strong performance within specific systems or operational settings, their generalization across water environmental systems remains weak. Models trained for a particular task in wastewater treatment

plants. For example, often fail to transfer to constructed wetlands or urban drainage systems without extensive retraining. This limitation reflects a deeper task-level issue: most AI models implicitly encode system-specific patterns rather than learning task-invariant representations.

Limited cross-system transferability constrains AI scalability and raises deployment costs. Future task-oriented research should focus on representation learning that disentangles task semantics from system-specific dynamics, enabling transfer learning, domain adaptation, and foundation-model reuse across diverse water environmental systems.

1) Future Directions: Improving cross-system transferability calls for representation learning strategies that disentangle task semantics from system-specific dynamics. Promising directions include domain adaptation, meta-learning, invariant representation learning, and foundation-model pretraining across diverse water environmental datasets. Future evaluation frameworks should explicitly incorporate cross-system transfer tests, few-shot adaptation scenarios, and robustness analysis under distribution shifts to ensure scalable and deployable AI solutions.

D. From Data-Driven Tasks to Physics-Informed Learning

A fundamental limitation underlying many task-level gaps is the reliance on purely data-driven learning paradigms, particularly in natural aquatic systems (NAS) and constructed wetlands (CWs). Unlike highly engineered systems, these water environmental systems are dominated by ecological processes, long-term dynamics, and strong spatiotemporal heterogeneity, where monitoring data are often sparse, noisy, and incomplete. As a result, task models trained solely on historical data frequently suffer from poor robustness and rapid performance degradation when deployed beyond their original operating conditions.

From a task-oriented perspective, this limitation reflects not only data scarcity but also the absence of physical and process constraints in task formulation. Predictive tasks may violate mass balance or transport principles, control-oriented tasks may generate physically infeasible actions, and knowledge discovery tasks may capture spurious correlations rather than causal relationships. These issues are particularly pronounced in NAS and CWs, where system evolution is governed by partially observable and weakly controllable natural processes.

Physics-informed machine learning (PIML), including physics-informed neural networks (PINNs) [119], provides a promising pathway to address these challenges by embedding governing equations, conservation laws, and mechanistic knowledge into task objectives or learning constraints [120]–[122]. By coupling data-driven learning with physical consistency, PIML improves generalization, interpretability, and robustness under limited data conditions. Consequently, physics-informed task formulations play a critical role in bridging data-driven intelligence and process-aware decision-making in water environmental systems.

1) Future Directions: Future task formulations should systematically integrate physical laws and mechanistic knowledge into AI training objectives and architectural design. Hybrid modeling frameworks that couple mechanistic simulators with neural networks, constraint-aware loss functions enforcing conservation principles, and uncertainty-aware physics-informed learning strategies represent promising pathways. Evaluation should explicitly assess physical consistency, conservation compliance, and robustness under sparse or noisy data conditions to ensure reliable deployment in complex environmental systems.

E. Emerging Trends Toward Unified and Knowledge Tasks

Unified task formulations represent an emerging direction in functional AI research for water environmental systems. Instead of addressing prediction, perception, and control as isolated objectives, recent studies increasingly investigate joint task formulations, particularly within reinforcement learning and multi-task learning frameworks. At the same time, the explicit integration of domain knowledge, physical constraints, and environmental regulations into task design is gaining importance, as it directly contributes to improved robustness, safety, and interpretability in real-world deployment scenarios.

Knowledge discovery is also undergoing a conceptual shift from a peripheral analytical tool toward a central task that actively informs model structure, task prioritization, and decision logic. When embedded into learning objectives or control policies, discovered knowledge enables AI systems to move beyond reactive responses and toward proactive, adaptive behavior in dynamic water environmental settings.

Recent advances in large language models (LLMs) and foundation models further extend this trend by introducing new mechanisms for knowledge-centric decision support. Through the integration of heterogeneous information sources, including monitoring data, operational records, environmental regulations, and technical documentation, LLM-based systems have the potential to support regulatory interpretation, scenario-based emergency planning, and cross-system reasoning. In this role, large models act as a high-level cognitive layer that coordinates perception, prediction, control, and knowledge tasks, thereby providing an enabling infrastructure for knowledge-aware and regulation-compliant AI-driven water environmental systems.

Even within emerging unified and knowledge centric task formulations, additional limitations become evident when water environmental systems are situated in the broader water–energy–food–climate nexus. Most existing task designs remain predominantly water-centric. Cross-domain interactions, including energy-intensive treatment processes, climate-driven hydrological variability, and water–food feedbacks, are rarely encoded into task objectives or learning constraints. As a result, predictive, control, and optimization tasks developed in isolation often fail to capture system-level trade-offs and cascading effects across coupled domains. This exposes a critical task-level gap. Unified and knowledge-driven AI tasks need to evolve beyond single-domain optimization toward cross-task formulations that explicitly account for nexus interactions.

1) Future Directions: Unified and knowledge-centric task design should evolve toward layered AI architectures in which foundation models coordinate task modules, knowledge graphs encode environmental regulations and domain con-

straints, and reinforcement learning enables adaptive decision-making across domains. Future research should also construct multi-domain benchmark environments incorporating water–energy–food–climate interactions to evaluate trade-offs, long-horizon stability, and regulatory compliance. Such developments will facilitate the transition from water-centric optimization toward nexus-aware intelligent infrastructure.

VII. Conclusion

This review does not treat water environmental intelligence as a collection of isolated application cases. Instead, it provides an integrated analysis across multiple water environmental systems and functional AI tasks, highlighting common methodological patterns as well as structural differences that are often less visible in system- or task-specific reviews. By jointly examining predictive modeling, pollution identification, intelligent control, multi-objective optimization, and knowledge discovery within urban drainage systems, wastewater treatment plants, natural aquatic systems, and constructed wetlands, this review illustrates how AI methods develop under different task demands and system characteristics.

From this cross-task and cross-system perspective, existing studies are shown to be unevenly distributed across tasks and systems, and interactions among tasks as well as methodological transfer across systems remain limited. Based on these observations, this review suggests that future research may benefit from moving beyond isolated task- or system-level designs toward more integrated and deployable AI frameworks that support coordinated analysis and decision making in practical water environmental systems.

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